GA1025 Macro Theory
Solution to Problem Set 1

Junbiao Chen*

September 11, 2025

1 Search Models

1.1 Exercise 7.1

A McCall Model with a chance of unemployment. The Bellman equation is as follows

$$v(w) = \max_{a \in \{\text{Accept, Reject}\}} \{ \frac{w}{1-\beta}, \beta(\phi(c + \bar{V}) + (1-\phi)\mathbb{E}\left[v(w')\right]) \} \tag{1.1}$$

where $\bar{V} = \frac{c+\beta(1-\phi)\mathbb{E}[v(w')]}{1-\beta\phi}$ denotes the continuation value of being unemployed.[1] $v(w)$ denotes the value function conditional on receiving an offer. $\mathbb{E}\left[v(w')\right] = \int_0^B v(w')dF(w')$ denotes the expected value conditional on receiving an offer.

1.2 Exercise 7.2

A McCall Model with two offers per period. Since $F(x) = \Pr(z \leq x)$, the cumulative distribution function of the maximum of two random variables i.i.d. drawn from $F(x)$ is $G(x_m) = \Pr(x < x_m) \cdot \Pr(x < x_m) = [F(x_m)]^2$.

$$v(w_m) = \max_{a \in \{\text{Accept, Reject}\}} \left(\frac{w_m}{1-\beta}, c + \beta\mathbb{E}_{\max}[v(w'_m)] \right) \tag{1.2}$$

*E-mail: jc14076@nyu.edu.

[1] I obtain $\bar{V}$ by solving the following Bellman equation: $\bar{V} = c + \beta\left(\phi\bar{V} + (1-\phi)\mathbb{E}\left[v(w')\right]\right)$

where $\mathbb{E}_{\max}[v(w'_m)] = \int_0^B v(w'_m)\, d[F(w'_m)]^2$. Since $\mathbb{E}_{\max}[v(w'_m)] = 2\int_0^B v(w'_m) f(w'_m) F(w'_m) dw' > \int_0^B v(w'_m) f(w'_m) dw'$,[2] we have

$$\bar{w}_{\max} = (1-\beta)\big(c + \beta\mathbb{E}_{\max}[v(w'_m)]\big) > (1-\beta)\big(c + \beta\mathbb{E}[v(w'_m)]\big) = \bar{w}.$$

■

1.3 Exercise 7.3

A McCall Model with n iid offers per period. Generalize the formulation in Exercise 7.2, we have

$$v(w) = \max_{a\in\{\text{Accept, Reject}\}} \left(\frac{w}{1-\beta}, c + \beta\mathbb{E}_{\max}[v(w')]\right) \tag{1.3}$$

where $\mathbb{E}_{\max}[v(w')] = n\int_0^B v(w') F^{n-1}(w') f(w') dw'$.

1.4 Exercise 7.4

A McCall Model with a random number of offers per period. Denote the probability of n in this period but m in the next period as π_{mn}.

Part a. The Bellman Equation is given as follows.

$$v(w,n) = \max_{\{\text{Accept, Reject}\}} \left(\frac{w}{1-\beta}, c + \beta(\mathbb{E}[v(w',n')])\right) \tag{1.4}$$

where w is the best offer. $\mathbb{E}[v(w',n')] = \sum_{m=1}^N \pi_{mn}\int_0^B mF(w')^{m-1} v(w', n'=m) dw'$ with $\sum_m \pi_{mn} = 1$.

Part b. We characterize the reservation wage $\bar{w}(n)$ as follows.

$$\bar{w}(n) = (1-\beta)\beta\left[\sum_{m=1}^N \pi_{mn}\int_0^B v(w', n'=m) dF(w')\right] + c(1-\beta) \tag{1.5}$$

where $\sum_{m=1}^N \pi_{mn}\int_0^B v(w', n'=m) dF(w') \equiv \mathbb{E}[v(w',m)]$ is the marginal expected value function given the number of offers arrived.

[2] We apply the monotonicity of $v(x)$ and $2F(x)-1$ to finish the proof:

$$\begin{aligned}
&2\int_0^B v(w'_m) f(w'_m) F(w'_m) dw' - \int_0^B v(w'_m) f(w'_m) dw' \\
=&\int_0^B v(w'_m)(2F(w'_m)-1) dF(w') \\
=&\mathbb{E}\left[v(w'_m)(2F(w'_m)-1)\right] \\
=&\underbrace{\text{Cov}\left[v(w'_m), 2F(w'_m)-1\right]}_{>0} + \underbrace{\mathbb{E}[v(w'_m)]\mathbb{E}[2F(w'_m)-1]}_{>0} > 0
\end{aligned}$$

Note that $\sum_m \pi_{mn}\mathbb{E}[v(w', m)]$ is a function of n, it follows that $\bar{w}(n)$ is also a function of n.

1.5 Exercise 7.5

A McCall Model with an endogenous number of offers per period. In this exercise we consider a case where the worker endogenously choose the number of offers $n \in 1, 2, \ldots, N$ before the wages are drawn. Formally, the Bellman Equation is given by:

$$v(w_m) = \max_{a \in \{\text{Accept, Reject}\}} \left(\frac{w_m}{1-\beta}, c + \beta \max_{n \in \{1,\ldots,N\}} \left[-k(n) + \int_0^B v(w'_m) d[F(w'_m)]^n \right] \right) \tag{1.6}$$

1.6 Exercise 7.7

A McCall Model with variable labor supply. We first show that the optimal labor supply is a time invariant policy function, which solely depends on w. Let $\tilde{v}(w)$ be the value function of accepting an offer at wage w. Since the state variable w does not change across periods, the Bellman equation that characterize the optimal labor supply is given by:

$$\tilde{v}(w) = \max_{l \in [0,1]} [u(w(1-l), l) + \beta \tilde{v}(w)] \tag{1.7}$$

Denote the optimal labor supply given wage w by $l^*(w)$, we have

$$u(w(1-l^*), l^*) = (1-\beta)\tilde{v}^*(w)$$

Therefore, the optimal labor supply $l^*(w)$ does not depend on time (because $\tilde{v}^*(w)$ is time-invariant.)[3]

Finally, we can specify the Bellman equation using the optimal labor supply policy $l^*(w)$ as follows.

$$v(w) = \max_{\{\text{Accept, Reject}\}} \left\{ \frac{u\Big(w(1-l^*(w)), l^*(w)\Big)}{1-\beta}, u(c,1) + \beta \int_0^B v(w') dF(w') \right\} \tag{1.8}$$

Note that the preservation wage structure in the Bellman equation above is given by following equation.

$$u\Big(w(1-l^*(w)), l^*(w)\Big) = (1-\beta)(u(c,1) + \beta \mathbb{E}[v(w')]) \tag{1.9}$$

1.7 Exercise 7.8

Wage growth rate and the reservation wage The Bellman Equation is given by

$$v(w) = \max_{\{\text{Accept, Reject}\}} \left\{ \frac{w}{1-\beta\phi}, c + \beta \int_0^B v(w') dF(w') \right\} \tag{1.10}$$

[3] Here I assume that the worker will stick to one optimal labor supply level under multiple equilibria.

Denote the reservation wage by $\bar{w}$, we have

$$\begin{aligned}\frac{\bar{w}}{1-\beta\phi} &= c+\beta\Big[\int_0^{\bar{w}} \frac{\bar{w}}{1-\beta\phi} dF(w') + \int_{\bar{w}}^{B} \frac{w'}{1-\beta\phi} dF(w')\Big] \\ &= c+\beta\Big[\frac{\bar{w}}{1-\beta\phi}\int_0^{B} dF(w') + \int_{\bar{w}}^{B} \frac{w'-\bar{w}}{1-\beta\phi} dF(w')\Big]\end{aligned} \tag{1.11}$$

$\Rightarrow$

$$\underbrace{\frac{(1-\beta)\bar{w}}{\beta} - \frac{c(1-\beta\phi)}{\beta}}_{g(\bar{w})} = \underbrace{\int_{\bar{w}}^{B} w' - \bar{w} dF(w')}_{h(\bar{w})} \tag{1.12}$$

We can use the graph below to visually show that $\frac{\partial \bar{w}}{\partial \phi} < 0$. (Because $g(w)$ shifts upward as ϕ increases, the intersection point moves leftward.)

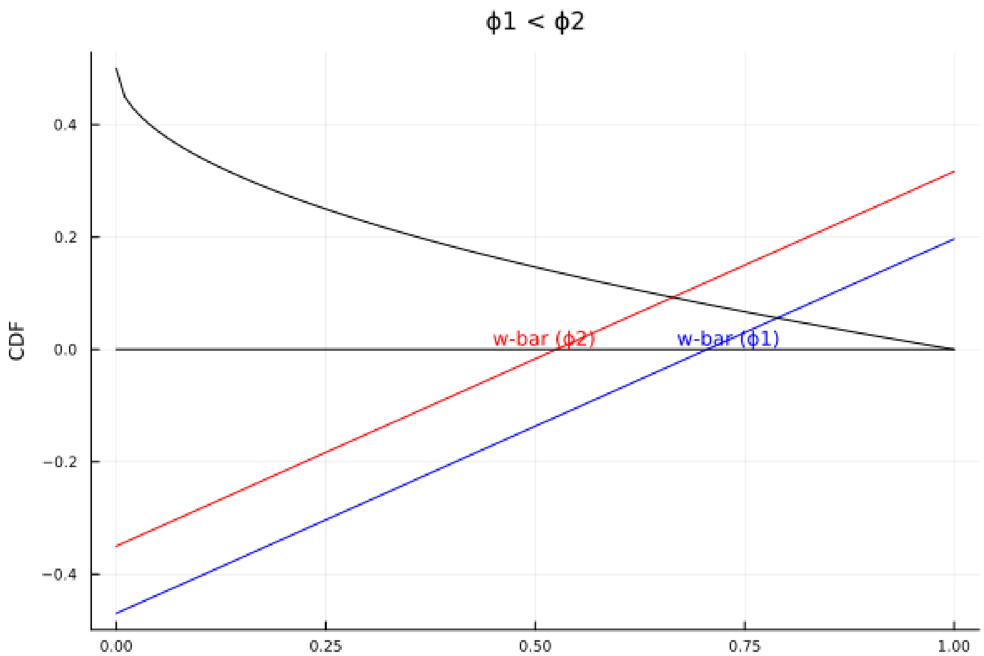

Figure 1.1: Wage growth and reservation wage

NOTES: The blue line and the red line represent functions $g(w)$ with ϕ_1 and ϕ_2, respectively, where $\phi_1 < \phi_2$.

1.8 Exercise 7.10

Finite horizon and mean-preserving spread. Denote the wage distribution by $F_1(w)$ and $F_2(w)$, where $F_1(w)$ is obtained from $F_2(w)$ using a mean-value spread. **Remark:** In this benchmark, there is no unemployment benefit.

(a) Since there is no unemployment benefit (UB), $\bar{w}_T = 0$ regardless of the wage distribution, the reservation wage at T conincide trivially. Now consider $t = T - 1$. The indifference condition without unemployment benefit is

$$\bar{w}_{T-1} = \frac{\beta(1-\beta^{T-t})}{1-\beta^{T-t+1}}\mathbb{E}[w']$$

, which doesn't depend on the wage distribution.

(b) According to the properties of MSP, if $F1$ is generated by $F2$ via MSP, then

$$\int_0^b F_2(w')dw' \leq \int_0^b F_1(w')dw', \forall b \in [0, B]$$

Therefore,

$$\begin{aligned}
\bar{w}_t|_{F_1} &= \frac{\beta(1-\beta^{T-t})}{1-\beta^{T-t+1}}\left[\mathbb{E}[w'] + \int_0^{\bar{w}_{t+1}} F_1(w')dw'\right] \\
\geq \bar{w}_t|_{F_2} &= \frac{\beta(1-\beta^{T-t})}{1-\beta^{T-t+1}}\left[\mathbb{E}[w'] + \int_0^{\bar{w}_{t+1}} F_2(w')dw'\right]
\end{aligned}$$

we complete the argument.

(c) **Now, we introduce the UB**. With UB, $\bar{w}_T = c$. As a result, the indifference condition at $T-1$ becomes

$$\bar{w}_{T-1} = c + \frac{\beta(1-\beta^{T-t})}{1-\beta^{T-t+1}}\left[\mathbb{E}[w'] - c + \int_0^c F(w')dw'\right]$$

, which is higher under F_1 following the argument in **(b)**.

2 Mean-preserving Spread

2.1 Answer to Question 2.1

(a) According to the definition of mean-perserving spread (MSP) in **?**, Y is a MSP from X if $Y = X + Z$ where $\mathbb{E}[Z|X] = 0$. Therefore,

$$\begin{aligned}
\text{Var}(Y) &= \text{Var}(X+Z) \\
&= \text{Var}(X+Z) \\
&= \text{Var}(X) + \text{Var}(Z) + 2\text{Cov}(X,Z) \\
&= \text{Var}(X) + \text{Var}(Z) + 2\Big(\mathbb{E}(XZ) - \mathbb{E}(X)\mathbb{E}(Z)\Big) \\
&= \text{Var}(X) + \text{Var}(Z) + 2\Big(\mathbb{E}(X\mathbb{E}(Z|X)) - \mathbb{E}(X)\mathbb{E}(\mathbb{E}[Z|X])\Big) \\
&= \text{Var}(X) + \text{Var}(Z) \\
&\geq \text{Var}(X)
\end{aligned}$$

■

(b) Y can be generated by X without using MSP while satisfies the properties of $\mathbb{E}(X) = \mathbb{E}(Y)$ and $\text{Var}(X) = \text{Var}(Y)$ as follows.

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim N\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}\right) \tag{2.1}$$

Remark: The process above is not a MSP because $\mathbb{E}[Z|X] \neq 0$.

Transitivity of mean-preserving spreads.

Note that $\mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)]$ if Y is a MSP from X for any concave utility function $u(\cdot)$. Suppose Y is a MSP from X, and Z is a MSP from Y. Then we have $\mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)] \geq \mathbb{E}[u(Z)]$. Therefore Z is a MSP from X, thus MSP is transitive. ■

(b) Denote $(F, G) \in S$ if distribution functions F and G satisfy the single-crossing property. The figure below shows that $(F1, F2) \in S$ and $(F2, F3) \in S$, but $(F1, F3) \notin S$. Visually, $F1$, and $F3$ cross twice.

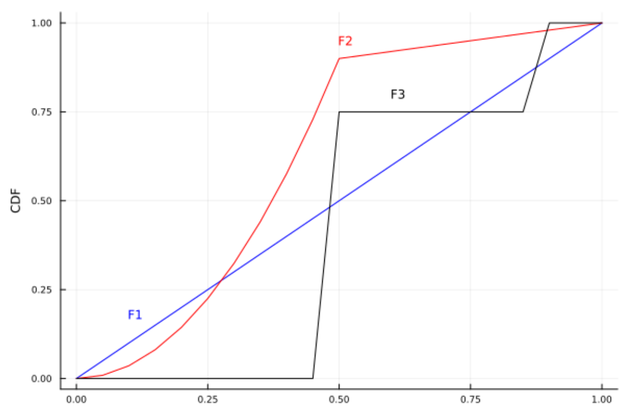

Figure 2.1: Single-crossing property is not transitive

3 Numerical implementation of the finite-horizon search problem

Answer to Question 3.1

The Bellman equation is given by

$$\begin{aligned} v_t(w) &= \max_{\{\text{Accept, Reject}\}} \left\{ \frac{w(1-\beta^{T-t+1})}{1-\beta}, c + \beta \int_0^B v_{t+1}(w')dF(w') \right\}, \quad \forall t = 0, 1, \dots, T \\ v_{T+1}(w) &= 0 \end{aligned} \tag{3.1}$$

Answer to Question 3.2

Since $v_{T+1}(w) = 0$, the maximization problem is given by

$$v_T(w) = \max_{\{\text{Accept, Reject}\}} \{w, c\}$$

Therefore,

$$v_T(w) = \max\{w, c\}$$

Answer to Question 3.3

We use induction to prove that the optimal policy is a reservation wage. When $t = T$, $\bar{w}_T = c$. When $t = T - 1$, the Bellman Equation is:

$$v_{T-1}(w) = \max_{\{\text{Accept, Reject}\}} \{w + \beta w, c + \beta(F(c)c + \int_c^B w' dF(w'))\}$$

Therefore, $\bar{w}_{T-1} = \frac{c+\beta(F(c)c+\int_c^B w' dF(w'))}{1+\beta}$, which is a constant.

Answer to Question 3.4

Inspired by the answers above, we can characterize the indifference condition as follows.

$$\frac{\bar{w}_t(1 - \beta^{T-t+1})}{1 - \beta} = c + \beta\left(F(\bar{w}_{t+1})c + \int_{\bar{w}_{t+1}}^B w' dF(w')\right)$$

Answer to Question 3.5

Yes, this problem has well-defined solution even if $\beta > 1$ as we can solve it using backward induction.

Answer to Question 3.6

Recall that the indifference condition for reservation wage is given by

$$\frac{\bar{w}_t(1 - \beta^{T-t+1})}{1 - \beta} = c + \beta \int_0^B v_{t+1}(w') dF(w') \tag{3.2}$$

Multiplying both sides with $1 - \beta$ and subtracting $(1 - \beta^{T-t+1})c$, we have

$$(\bar{w}_t - c)(1 - \beta^{T-t+1}) = (\beta^{T-t+1} - \beta)c + \beta(1 - \beta) \int_0^B v_{t+1}(w') dF(w')$$

Recall that

$$v_{t+1}(w') = \begin{cases} \frac{\bar{w}_{t+1}(1-\beta^{T-t})}{1-\beta} & \text{if } w' \leq \bar{w}_{t+1}, \\ \frac{w'(1-\beta^{T-t})}{1-\beta} & \text{if } w' > \bar{w}_{t+1} \end{cases}$$

Therefore, we have

$$
\begin{aligned}
(\bar{w}_t - c)(1-\beta^{T-t+1}) &= -\beta(1-\beta^{T-t})c + \beta(1-\beta^{T-t})\bar{w}_{t+1}\int_0^{\bar{w}_{t+1}} dF(w') + \beta(1-\beta^{T-t})\int_{\bar{w}_{t+1}}^{B} w'dF(w') \\
&= -\beta(1-\beta^{T-t})c + \beta(1-\beta^{T-t})\int_0^{\bar{w}_{t+1}} (\bar{w}_{t+1}-w')dF(w') + \beta(1-\beta^{T-t})\mathbb{E}[w'] \\
&= \beta(1-\beta^{T-t})[\mathbb{E}[w'] - c + \int_0^{\bar{w}_{t+1}} (\bar{w}_{t+1}-w')dF(w')]
\end{aligned}
$$

Applying Integral by Part, $\int_0^{\bar{w}_{t+1}}(\bar{w}_{t+1}-w')dF(w') = \int_0^{\bar{w}_{t+1}} F(w')dw'$. Therefore, we have

$$
\bar{w}_t = c + \frac{\beta(1-\beta^{T-t})}{1-\beta^{T-t+1}}\left[\mathbb{E}[w'] - c + \int_0^{\bar{w}_{t+1}} F(w')dw'\right]
$$

Now denote $\frac{\beta(1-\beta^{T-t})}{1-\beta^{T-t+1}} = \tilde{\beta}_{t+1}$, and $g_{t+1}(w) = c + \tilde{\beta}_{t+1}\left[\mathbb{E}[w'] - c + \int_0^{w} F(w')dw'\right]$. Without loss of generality, I assume $\mathbb{E}[w'] > c$. It follows that

- $g'_{t+1}(w) = \tilde{\beta}_{t+1}F(w) \in [0,1)$
- $g''_{t+1}(w) = \tilde{\beta}_{t+1}f(w) > 0$
- Fix w_o, $g_{t+1}(w_o) - g_t(w_o) = \left[\frac{(1-\beta^{T-t-1})\beta}{1-\beta^{T-t}} - \frac{(1-\beta^{T-t})\beta}{1-\beta^{T-t+1}}\right]\left[\mathbb{E}[w'] - c + \int_0^{w_o} F(w')dw'\right] < 0$.

To complete the proof, we start with $t = T-1$, and inductively show that $\bar{w}_t < \bar{w}_{t+1}$ using the following figure. When $t = T-1$, $\bar{w}_T = c$, and thus, $g_T(c) = c + \frac{\beta}{1+\beta}\left[\mathbb{E}[w'] - c + \int_0^{c} F(w')dw'\right] > c$.

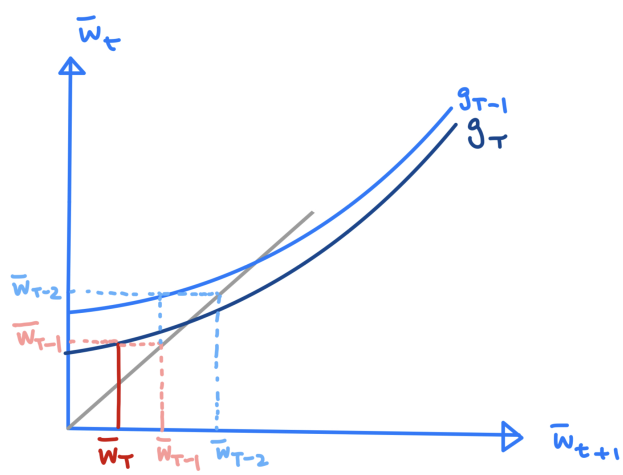

Figure 3.1: Using induction to show $\bar{w}_t < \bar{w}_{t+1}$

Answer to Question 3.7

Fix t, we have

$$v_t(w') = \begin{cases} \frac{\bar{w}_t}{1-\beta} & \text{if } w' \leq \bar{w}_t \\ \\ \frac{w'}{1-\beta} & \text{if } w' > \bar{w}_t \end{cases}$$

as $T \to \infty$. In follows that, the indifference condition for reservation wage is given by

$$\bar{w}_t - c = \beta \left[\mathbb{E}[w'] - c + \int_0^{\bar{w}_{t+1}} F(w')dw' \right]$$

We can guess and verify that $\bar{w}_t = \bar{w}_{t+1} = \bar{w}$ and

$$v_t(w) = \begin{cases} \frac{\bar{w}}{1-\beta} & \text{if } w \leq \bar{w} \\ \\ \frac{w}{1-\beta} & \text{if } w > \bar{w} \end{cases}$$

as $T \to \infty$.

Quantitative Exercise

We now solve the model numerically in two different ways. In what follows, use the following benchmark parameter values: $T = 50$, $\beta = 0.97$, $B = 1$, $c = 0.3$. For the distribution $F(w)$ of wage offers, start with $w \sim \text{Beta}(\alpha, \beta)$, with parameters $\alpha = 2, \beta = 2$. Recall that the Beta (α, β) distribution has the density

$$f(w) = \frac{w^{\alpha-1}(1-w)^{\beta-1}}{\mathrm{B}(\alpha, \beta)},$$

where $\mathrm{B}(\alpha, \beta)$ is the Beta function.

My code is available at:

https://github.com/BillJunbiaoChen/GA1025_MacroTheory/tree/main/ProblemSets/PS1_computation

Answer to Question 3.8

I discretize w using 101 evenly spaced grids $\{w^j\}_{j=1}^{101}$ on $[0, B]$, and apply the following scheme to obtain probabilities $\{\tilde{f}_j\}_{j=1}^{101}$:

$$\tilde{f}^j = \begin{cases} F\left(\frac{1}{2}\left(\tilde{w}^2 + \tilde{w}^1\right)\right), & j = 1 \\ \\ F\left(\frac{1}{2}\left(\tilde{w}^{j+1} + \tilde{w}^j\right)\right) - F\left(\frac{1}{2}\left(\tilde{w}^j + \tilde{w}^{j-1}\right)\right), & 1 < j < J \\ \\ 1 - F\left(\frac{1}{2}\left(\tilde{w}^J + \tilde{w}^{J-1}\right)\right), & j = J \end{cases}$$

Thus, the discretized version of the Bellman equation is given by

$$\hat{V}_t^j = \max_{\{\text{Accept, Reject}\}} \left\{ \frac{w^j(1-\beta^{T-t+1})}{1-\beta} + \beta E[\hat{V}_{t+1}^j] \right\}, \quad \forall t = 0, 1, \ldots, T$$

where expected value function at $t+1$ is given by

$$E[\hat{V}_{t+1}^j] = \sum_{\{i: w_i < \bar{w}_{t+1}\}} \bar{w}_{t+1} f_i + \sum_{\{i: w^i \geq \bar{w}_{t+1}\}} w^i f_i$$

Answer to Question 3.9

The figure below plots the reservation wages against time.

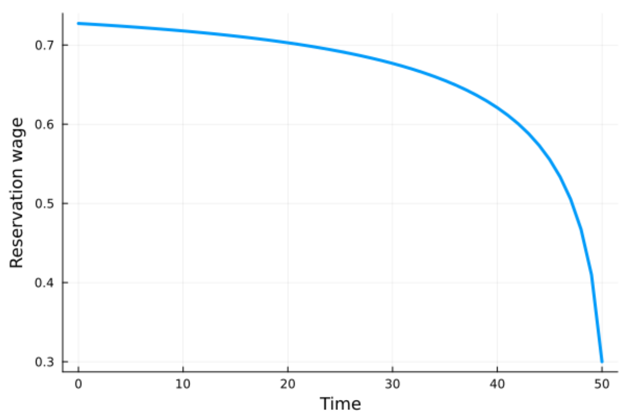

Figure 3.2: Value functions and reservation wages

NOTES: Value functions from $t = 1, \ldots, T$ are depicted from up to down.

Answer to Question 3.10

By iterating on the indifference condition, I obtain the same reservation wages as above. The figure below plots the value functions against time.

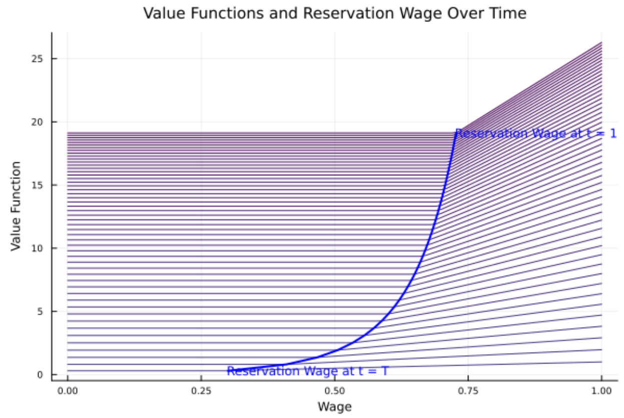

Figure 3.3: Value functions and reservation wages

NOTES: Value functions from $t = 1, \ldots, T$ are depicted from up to down.

Answer to Question 3.11

Using the specification with $w \sim \text{Beta}(1.5, 1.5)$, I obtain the reservation wage and value functions as follows. Intuitively, the reservation wage increases after MSP as the continuation value for waiting for better offers increases. Note that the realization below the observation wage doesn't affect the value function. Therefore, after MSP, the worker can expect better offers. As predicted by the

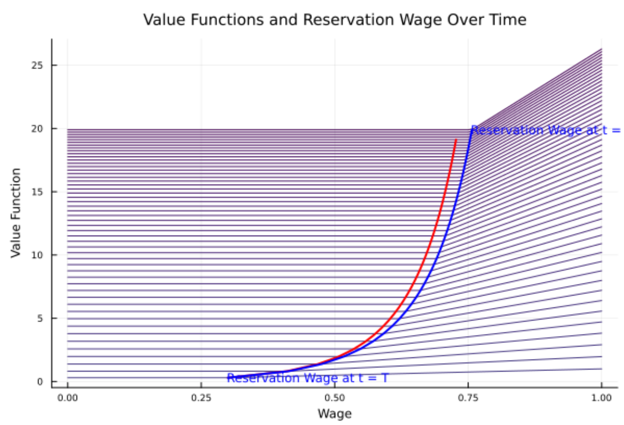

Figure 3.4: Value functions and reservation wages

NOTES: Value functions from $t = 1, \ldots, T$ are depicted from up to down. **Reservation wage** at $t = 1$ after MSP is 0.75689. The original reservation wage curve is depicted in red, while the one with MSP is depicted in blue.

model, the reservation wage increases after MSP as the continuation value for waiting for better offers increases.

Answer to Question 3.12

Recall that the indifference condition for the reservation wage in the infinite horizon is given by

$$\begin{aligned}\frac{\bar{w}}{1-\beta} &= c+\beta\left[\int_0^{\bar{w}} \frac{\bar{w}}{1-\beta}dF(w') + \int_{\bar{w}}^{B} \frac{w'}{1-\beta}dF(w')\right] \\ &= c+\beta\left[\frac{\bar{w}}{1-\beta} + \int_{\bar{w}}^{B} \frac{w'-\bar{w}}{1-\beta}dF(w')\right]\end{aligned}$$

$\Rightarrow$

$$\bar{w}-c = \frac{\beta}{1-\beta}\int_{\bar{w}}^{B}(w'-\bar{w})dF(w')$$

Define a function $g(w) = w-c-\frac{\beta}{1-\beta}\int_w^B(w'-w)dF(w')$, and then apply discretization listed above, we have

$$\tilde{g}(w) = w-c-\frac{\beta}{1-\beta}\sum_{\{i:w_i>w\}}(w_i-w)f_i \tag{3.3}$$

To solve the equation above, I apply `Roots` package in `Julia`. The table compares the reservation wages (i.e., the root in equation **??**) obtained via different algorithms and box constraints.

Interval	Algorithm	Time (s)	Root
Parameterization: $\beta = \alpha = 2$			
$[0, 1]$	Secant	0.020007	0.74870
$[0, 1]$	Bisection	0.029884	0.74870
$[0, 1]$	A42	0.027841	0.74870
$[0.3, 1]$	Secant	0.006241	0.74870
$[0.3, 1]$	Bisection	0.008908	0.74870
$[0.3, 1]$	A42	0.009308	0.74870
Parameterization: $\beta = \alpha = 1.5$			
$[0, 1]$	Secant	0.225221	0.77908
$[0, 1]$	Bisection	0.095937	0.77908
$[0, 1]$	A42	0.310508	0.77908
$[0.3, 1]$	Secant	0.014162	0.77908
$[0.3, 1]$	Bisection	0.016413	0.77908
$[0.3, 1]$	A42	0.015981	0.77908

Table 3.1: Reservation wage in infinite horizon computed by root-finding algorithms.

Next I apply numerical experiments to show that the reservation wage in infite horizon equals to $\lim_{T\to\infty}\bar{w}_1$. Concretely, I experiment with $T = 50, 51, 52, 53, 54, 55, 60, 65, 75, 100, 200, 300$, and plot the reservation wages at $t = 1$ across different T. The figure indicates that $\lim_{T\to\infty}\bar{w}_1 = \bar{w}$.

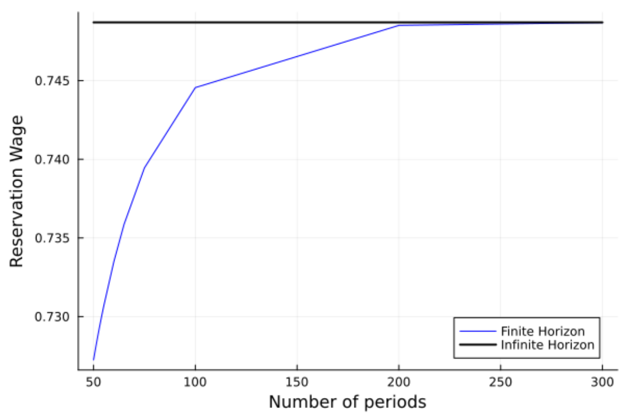

Figure 3.5: The convergence of reservation wages